

Elif Akata

PhD student @ Human-Centered AI Lab, Helmholtz Munich & University of Tübingen

✉ elif.akata@uni-tuebingen.de 🏠 eliaka.github.io 📄 google scholar

Education

- University of Tübingen** 10/2022 – present
PhD Machine Learning & Cognitive Science Munich, Germany
- LLMs as social and collaborative agents, human-AI coordination
 - Supervisors: Eric Schulz, Matthias Bethge
 - International Max Planck Research School for Intelligent Systems (IMPRS-IS) Scholar
- University of Tübingen** 10/2019 – 2/2022
MSc Computer Science Tübingen, Germany
- Thesis on attention detection in the classroom (Human-Centered Technologies for Learning Lab, supervised by Enkelejda Kasneci)
- Saarland University** 10/2015 – 9/2019
BSc Computer Science Saarbrücken, Germany
- Thesis on EMG-based hand gesture recognition (Human-Computer Interaction Lab, supervised by Jürgen Steimle)

Experience

- Research Scientist Intern** New York, USA
Meta FAIR 7/2026 - present
- Multi-Agent Multimodal Foundations
- Visiting Researcher** Oxford, UK
University of Oxford, British Open-Ended Learning & Discovery Lab 1/2026 – 6/2026
- Human-Centric Learning, supervised by Jakob Foerster
- Research Assistant** Tübingen, Germany
Tübingen AI Center, Computational Neuroscience and Machine Learning Lab 10/2021 – 9/2022
- Developed computational creativity and generative art applications by programming the Boston Dynamics Spot robot with the Spot SDK to autonomously generate drawings.
- Research Assistant** Tübingen, Germany
University of Tübingen, Human-Centered Technologies for Learning Lab 8/2021 – 9/2021
- Processed and analysed classroom activity data for an attention-based automated detection system.
- Research Intern** Stuttgart, Germany
Max Planck Institute for Intelligent Systems, Haptic Intelligence 9/2020 – 11/2020
- Contributed to the sensor-specific component of an ASL sign-recognition device translating hand movements into text, supervised by Katherine Kuchenbecker
- Student Research Assistant** Saarbrücken, Germany
Max Planck Institute for Software Systems, Programming Analysis and Verification 5/2017 – 4/2019
- Built benchmark programs for the Daisy numerical optimization framework and evaluated performance on resource-constrained hardware like Arduinos, supervised by Eva Darulova

Publications

- **Akata, E.**, Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., and Schulz, E. (2025). Playing repeated games with large language models. *Nature Human Behaviour*.
- **Akata, E.***, Voudouris, K.*, Fortuin, V., and Schulz, E. (2026). In-context function learning in large language models. *AISTATS*.
- Buschhoff, L. M. S.*, **Akata, E.***, Bethge, M., and Schulz, E. (2025a). Visual cognition in multimodal large

language models. *Nature Machine Intelligence*.

- Binz, M., **Akata, E.**, Bethge, M., et al. (2025). A foundation model to predict and capture human cognition. *Nature*.
- Buschhoff, L. M. S.*, Voudouris, K.*, **Akata, E.**, Bethge, M., Tenenbaum, J. B., and Schulz, E. (2025b). Testing the limits of fine-tuning to improve reasoning in vision language models. *ICML*.

Learning and Teaching

Brains, Minds & Machines Summer Course

MIT CBMM, Participant

Aug 2024

Woods Hole, USA

International Computer Vision Summer School (ICVSS)

University of Catania & University of Cambridge, Participant

Jul 2023

Sicily, Italy

Trustworthy Machine Learning

TA for a 6 Credit MSc lecture at University of Tübingen

Winter 2022/2023

Tübingen, Germany

Supervision

Irais Garces de Marcilla Lappin | MSc thesis | Max Planck Institute of Psychiatry | 2026

Hanru Zhang | MSc thesis | University of Tübingen | 2024

Invited Talks

Generative AI and (Decision) Neuroscience Symposium

Neuro2026, annual meeting of the Japan Neuroscience Society

Aug 2026

Kobe, Japan

Center for Information and Neural Networks (CiNet)

Osaka University, hosted by Masahiko Haruno

Jul 2026

Osaka, Japan

Trustworthy AI Lab Talk

UCLA, hosted by Guang Cheng

Jun 2026

Los Angeles, USA

Human Information Processing Lab Talk

University of Oxford, hosted by Chris Summerfield

Feb 2026

Oxford, UK

Algorithmic Game Theory, Guest Lecturer

Hertie School, hosted by Asya Magazinnik

Nov 2025

Berlin, Germany

Generative AI in Social Science Research Workshop

London School of Economics, hosted by the Department of Methodology

Jun 2024

London, UK

Experimental Psychology Departmental Talk

UCL, hosted by Maarten Speekenbrink

Jun 2024

London, UK

Awards

BMM 2024 Summer School Travel Grant & Scholarship

MIT Center for Brains, Minds and Machines, \$3000

2024

Cyber Valley Start-up Incubation Program

Our startup concept won the Sustainability Award sponsored by Carl Zeiss Foundation, €2000

2023

Skills and Interests

Languages: English (IELTS Academic Band 7,5), German (DSH-2, CEFR Level C1), Turkish (Native)

Coding: Python, C/C++, PyTorch, Numpy, SciKit-Learn, Git, model fine-tuning (SFT, GRPO) on HPC

Drawing & Illustration (my design for our paper made the cover of Nature Machine Intelligence in January 2025)

Tennis (club player now and was decent as a junior with Edirne U12 #1 and Marmara U10 #3 rankings)